

SAKLAIN NIAM

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Education

BSc (Engineering) in Food Engineering and Tea Technology 2019-2024
Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh
Thesis: Impact of alginate, guar gum, and pectin-based edible coatings on strawberry shelf life.
CGPA: 3.46/4.00

Professional Experience

Research Assistant 5/2024 - Present
Department of Food Engineering and Tea Technology, SUST
Supervisor: Dr. Iftekhar Ahmad

Selected Publications

- **Niam, S.**, Ahmad, I., Rayhan, M. A., Mahmood, S., Jon, P. H., & Ahmed, M. M. (2025). Machine learning-based optimization of alginate, guar gum, and pectin-based edible coatings for extended strawberry shelf life. *LWT*, 233, 118548. <https://doi.org/10.1016/J.LWT.2025.118548>
- Dina, P. R., **Niam, S.**, Ahmad, I., Zzaman, W., Salim, M., Ahmed, Md. M., Rahman, T., & Shifat, A. S. (2026). Comprehensive Analysis of Tea Concentrates with Machine Learning-Assisted Sensory Characterization. (Applied Food Research). <https://doi.org/10.2139/SSRN.5398137>
- Application of Computer Vision for Classification of Bread Made from All-Purpose and Multigrain Flour with Physicochemical Characterization (manuscript submitting soon).

Conference Papers

- **Niam, S.**, Rahman, T., Patwary, A., Hossain, M. (2025). A machine learning framework integrating seed traits and plasma parameters for predicting germination uplift in crops. *International Plant Tissue Culture and Biotechnology Conference* <https://arxiv.org/abs/2510.23657v1>
- S. M., S. M., Iftekhar, A., Zzaman, W., Patwary, M. A., Jon, P. H., & **Niam, S.** (2025). Response surface optimization of the extraction of bioactive compounds in CTC black tea infusion. International Tea Symposium, Sri Lanka.

Manuscript Preparation (Ongoing)

- Real-time UAV-based Tomato Plant Counting and Weed Mapping for Small-Scale Agricultural Fields
- Accurate Instance Re-Identification and Mapping of Cauliflower Plants Using Drones

Projects

- Ground Robot-Based Lettuce Phenotyping: Segmentation, Counting, and Trajectory Tracking - [Link](#)
- Real Time Apple Tracking using BotSORT and LoFTR in Apple Orchard - [Link](#)
- Small-scale weed detection and segmentation from drone imagery using various Faster RCNN and mask RCNN with RPN and FPN architecture - [Link](#)
- Tea Leaf Disease classification - [Link](#)
- MIFlow experimental tracker for seed germination rate prediction - [Link](#)

- Real-time activity monitoring system for tea plantation workers with YOLOv8
- Cheminformatics in Python: Predicting Solubility of Molecules - [Link](#)
- Bioactivity of Drugs Prediction from Chemical Structures - [Link](#)
- Climatic analysis of the August 2024 flood in Bangladesh
- Plant Layout Optimization for PinePlus Probiotic Pineapple Juice Manufacturing
- Thermal Behaviour of Protein - Flavonoid Complex: a Qualitative Study - [Link](#)

Technical Skills

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|---|--|
| • Programming: Python, LaTeX | • Lab Instruments: GC, HPLC, Spectrophotometry, E-Nose |
| • Machine Learning & Neural Networks | • Bioinformatics: chEMBL, RDKit |
| • Reinforcement Learning: DQN | • Statistics: Variance tests, PCA, LDA |
| • Computer Vision: OpenCV, Transfer Learning, Vision Transformers | • Scientific Software: Statistica, Minitab, Super-Pro, OriginPro |
| • Geostatistics and Satellite Data: Kriging (pyKrig), Causal Analysis (PCMCI), Heatwave Analysis | • Design & Media: Photoshop, Premiere Pro, Canva |
| • Wet Lab: Chemical & Microbiological Analysis, Bioactives Extraction, Thermal/non-Thermal Processing | • Audio Signal Analysis: Ableton |
| | • Microsoft Office |

Research Interests

- Physics-informed machine learning/neural network
- High-throughput phenotyping, yield analysis, hyperspectral imaging, and sensors in agriculture
- Precise irrigation and pesticides application using UAVs
- Robotics, Digital twins
- Remote sensing, Geospatial Analysis
- Application of non-thermal technologies to improve seed dormancy and bioactive extraction
- Post-harvest preservation and processing
- Improving land conservation and policymaking

Test Scores

- IELTS: Overall 7 — Reading 8 — Listening 7.5 — Speaking 7 — Writing 6
- GRE: Overall 306 — Quant 162 — Verbal 144 — Analytical 3

Trainings

Training in Quality Control and Unit Operation in Food Industry	01/2023 - 02/2023
Training Institute for Chemical Industries (TICI), Narshindi, Dhaka, Bangladesh	
Agro-Processing, Preservation, and Marketing Workshop	11/2022
Bangladesh Rice Research Institute (BRRI), Dhaka, Bangladesh	

Development and Industrialization of International Tea Trade for Belt and Road Countries

11/2022

Zhangzhou College of Science and Technology, Fujian, China

Extracurricular Activities

Shikorh Cultural Club

Publication Secretary

2023 - 2024

Saifurs Kulaura

IELTS Instructor

1/2025 - 3/2025

References

Dr. Iftekhar Ahmad

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Md. Amjad Patwary

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